EIGENVALUES OF 6×6 DOUBLY STOCHASTIC MATRICES WITH CONVEXLY INDEPENDENT EIGENVECTORS

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ABSTRACT. Levick, Pereira and Kribs conjectured that if an eigenvalue λ of an $n \times n$ doubly stochastic matrix has an eigenvector whose entries are convexly independent, then λ lies in the Perfect–Mirsky region $\bigcup_{k \le n} \Pi_k$, where Π_k is the convex hull of the k-th roots of unity. We prove the stronger inclusion $\lambda \in \Pi_6$ for $n = 6$; consequently, the eigenvalues of 6×6 doubly stochastic matrices that admit a convexly independent eigenvector form exactly the regular hexagon Π_6. The proof splits according to $\arg \lambda$. The sector $|\arg \lambda| \le \pi/3$ follows from a classical bound of Dmitriev and Dynkin. For $|\arg \lambda| \ge 2\pi/3$ we use that the convex hull of the 3-element subset sums of the eigenvector is centrally symmetric, which reduces the problem to real 3×3 matrices whose columns have ℓ^1-norm at most 1. The middle sector reduces to a statement about hexagons formed by two interleaved triangles with centroid at the origin. This statement is sharp along a two-parameter family of hexagons; we prove it by combining three exact "potential" lemmas with a branch-and-bound computation whose 53,093 boxes are verified in ball arithmetic. Bouthat, Mashreghi and Morneau-Guérin have announced, in talks, the inclusion $\lambda \in \Pi_n$ for all n; the argument given here for $n = 6$ is independent of theirs.

1. INTRODUCTION

A real square matrix is *doubly stochastic* if its entries are nonnegative and all its row and column sums are equal to 1. Let $\mathcal{D}_n$ denote the set of $n \times n$ doubly stochastic matrices and

$$\omega_n = \{\lambda \in \mathbb{C} : \ \lambda \text{ is an eigenvalue of some } A \in \mathcal{D}_n\}.$$

For $k \ge 1$ let $\Pi_k = \operatorname{conv}\{e^{2\pi i j/k} : 0 \le j < k\}$ be the convex hull of the k-th roots of unity (so $\Pi_1 = \{1\}$ and $\Pi_2 = [-1, 1]$), and let $\mathrm{PM}_n = \bigcup_{k=1}^{n} \Pi_k$. Taking convex combinations of the powers of the permutation matrix of a k-cycle ($k \le n$) shows that $\Pi_k \subseteq \omega_n$, so $\mathrm{PM}_n \subseteq \omega_n$, and Perfect and Mirsky [15] conjectured that $\omega_n = \mathrm{PM}_n$. The conjecture holds for $n \le 3$ [15] and for $n = 4$ [12]; it fails for $n = 5$ [13] (see [7] for a detailed computational study of the exceptional region); and it is open for every $n \ge 6$. Harlev, Johnson and Lim [7] formulate a conjecture on the boundary of ω_n and give computational evidence that, if it holds, then $\omega_n = \mathrm{PM}_n$ for $6 \le n \le 11$.

After settling the case $n = 4$, Levick, Pereira and Kribs [12] proposed three conjectures. Conjectures 1 and 3 of [12] were disproved for all $n \ge 5$ by Kim and Kim [11]. Following [12, Definition 2], the entries of a vector $v \in \mathbb{C}^n$ are *convexly independent* if no v_i lies in the convex hull of the other entries; equivalently, $v_1, \dots, v_n$ are the n vertices of a convex n-gon.

> **Conjecture 2 of** [12]**.** Let $A \in \mathcal{D}_n$ and let λ be an eigenvalue of A. If the entries of an eigenvector corresponding to λ are convexly independent, then $\lambda \in \mathrm{PM}_n$.

Our main result settles this conjecture for $n = 6$, in the stronger form $\lambda \in \Pi_6$.

Theorem 1.1. *Let $A \in \mathcal{D}_6$ and let λ be an eigenvalue of A. If λ has a right or a left eigenvector whose six entries are convexly independent, then $\lambda \in \Pi_6$.*

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The inclusion is sharp: if P is the permutation matrix of a 6-cycle, then $(\zeta^j)_{j=1}^6$ with $\zeta = e^{i\pi/3}$ is an eigenvector of every matrix $\sum_{l=0}^5 c_l P^l$ ($c_l \geq 0$, $\sum c_l = 1$), its entries form a regular hexagon, and the corresponding eigenvalues fill Π_6. Hence the set of eigenvalues of matrices in $\mathcal{D}_6$ admitting a convexly independent (left or right) eigenvector is exactly Π_6. Theorem 1.1 also says where a counterexample to the Perfect–Mirsky conjecture for $n = 6$ cannot come from:

Corollary 1.2. *If $\lambda \in \omega_6 \setminus \mathrm{PM}_6$, then no left or right eigenvector of any $A \in \mathcal{D}_6$ for the eigenvalue λ has convexly independent entries.*

Relation to the work of Bouthat, Mashreghi and Morneau-Guérin. In two talks given in May 2026 [3, 2] (see also the poster [1]), Bouthat, Mashreghi and Morneau-Guérin announced a theorem stating that, for every n, an eigenvalue of an $n \times n$ doubly stochastic matrix with a convexly independent eigenvector lies in Π_n. Theorem 1.1 is therefore the case $n = 6$ of their announced result. To our knowledge their paper is not yet publicly available. As far as we can tell from the slides, their approach considers, in each direction, an eigenpair of maximal modulus with a convexly independent eigenvector z, and aims to show that the vector of midpoints $\left(\frac{z_1+z_2}{2}, \ldots, \frac{z_n+z_1}{2}\right)$ is again a maximal eigenvector; iterating, one reaches affinely regular polygons, on which $|\lambda|$ can be maximized directly. This maximality-preservation step is labelled as a conjecture in the slides of the ILAS talk [2] and in one version of the Delaware slides [3]; another version of the Delaware slides, the file currently attached to the r-libre record of that talk [4], labels it a proposition, without proof. We have not seen a written proof of this step. The proof given here follows a different route and does not depend on that step. It is specific to $n = 6$, it uses only $\sum_i v_i = 0$ and the inclusions $\lambda K_m(v) \subseteq K_m(v)$ for $m = 1, 2, 3$, where $K_m(v)$ is the convex hull of the sums of m entries of v with distinct indices (Section 2), and part of it is computer-assisted.

Outline of the proof. Let $Av = \lambda v$ with $\lambda \notin \mathbb{R}$, $\operatorname{Im}\lambda > 0$, and assume that the entries of v are convexly independent. For $1 \leq m \leq 5$ let $K_m(v)$ be the convex hull of the sums $\sum_{i\in S} v_i$ over all m-element sets $S \subseteq \{1, \ldots, 6\}$. Then $\sum_i v_i = 0$ and $\lambda K_m(v) \subseteq K_m(v)$ for every m (Lemma 2.1). We distinguish three sectors (Figure 1).

(1) $0 < \arg\lambda \leq \pi/3$. Since ω_6 is contained in the Karpelevič region Θ_6 (the set of eigenvalues of 6×6 stochastic matrices), a classical bound of Dmitriev and Dynkin gives $\lambda \in \Pi_6$ (Theorem 2.4). Convex independence is not needed here.

(2) $2\pi/3 \leq \arg\lambda < \pi$. Because $n = 6$ is even and $\sum_i v_i = 0$, the polygon $K_3(v)$ is centrally symmetric, and under convex independence it is a hexagon. The inclusion $\lambda K_3(v) \subseteq K_3(v)$ then shows that λ is an eigenvalue of a real 3×3 matrix whose columns have ℓ^1-norm at most 1, and such eigenvalues lie in Θ_6. Applying the Dmitriev–Dynkin bound to $-\bar\lambda$ gives $\lambda \in \Pi_6$ (Proposition 3.2). This is the analogue of the argument of [12] for $n = 4$, where $K_2(v)$ is a parallelogram.

(3) $\pi/3 < \arg\lambda < 2\pi/3$. Here $\lambda \in \Pi_6$ means $\operatorname{Im}\lambda \leq \sqrt{3}/2$. The hexagon $H = K_2(v)$ has vertices $s_j = v_j + v_{j+1}$, and the triangles $s_1s_3s_5$ and $s_2s_4s_6$ both have centroid 0. We show that every such hexagon contains a point $x \neq 0$ with $\operatorname{Im}(s_j/x) \leq \sqrt{3}/2$ for all j (Proposition 4.3, which we call P_{90}); since $\lambda x \in H$, this forces $\operatorname{Im}\lambda \leq \sqrt{3}/2$ (Lemma 4.2).

Proposition P_{90} is the heart of the matter. It is sharp along a two-parameter family of hexagons (Remark 4.4), so it cannot be verified by a direct interval computation of its margin. We prove three exact lemmas (Section 5), each of which produces a witness vertex provided one particular inequality holds, and we use a branch-and-bound computation in ball arithmetic to check that, everywhere in a compact 6-dimensional box of parameters, one of finitely many such certificates applies (Section 6).

Notation. For $a, b \in \mathbb{C}$ we write $\mathrm{cr}(a, b) = \operatorname{Im}(\bar a b)$ and $\mathrm{dt}(a, b) = \operatorname{Re}(\bar a b)$ for the cross and dot products of a, b viewed as vectors in $\mathbb{R}^2$. Throughout, $\zeta = e^{i\pi/3}$ and $\omega = \zeta^2 = e^{2\pi i/3}$.

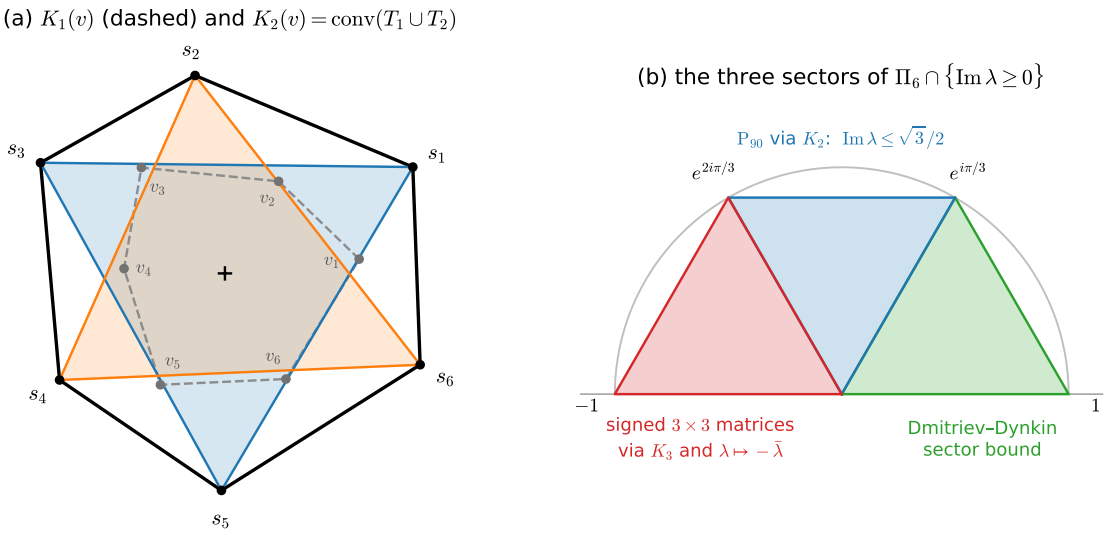


FIGURE 1. (a) A convexly independent vector v with $\sum_i v_i = 0$; the hexagon $K_2(v)$ has the vertices $s_j = v_j + v_{j+1}$, and $T_1 = s_1 s_3 s_5$, $T_2 = s_2 s_4 s_6$ are triangles with centroid 0. (b) The three sectors of the proof of Theorem 1.1: Theorem 2.4 for $|\arg\lambda| \leq \pi/3$, Proposition 4.3 with Lemma 4.2 for $\pi/3 < |\arg\lambda| < 2\pi/3$, and Proposition 3.2 for $|\arg\lambda| \geq 2\pi/3$.

Indices of vertices of hexagons are taken modulo 6. For a compact convex set $H \subset \mathbb{C}$ we write $S(H) = \{\lambda \in \mathbb{C} : \lambda H \subseteq H\}$ and $h_H(\varphi) = \max_{z \in H} \mathrm{Re}(z e^{-i\varphi})$ for its support function.

2. Preliminaries

2.1. **Subset-sum polygons.** For $v \in \mathbb{C}^n$ and $1 \leq m \leq n-1$ let

$$K_m(v) = \mathrm{conv}\Big\{\sum_{i \in S} v_i : \ S \subseteq \{1, \dots, n\}, \ |S| = m\Big\}.$$

Lemma 2.1. *Let $A \in \mathcal{D}_n$, $\lambda \in \mathbb{C} \setminus \{1\}$ and $v \in \mathbb{C}^n \setminus \{0\}$ with $Av = \lambda v$. Then*

(a) $\sum_{i=1}^n v_i = 0$;
(b) $\lambda K_m(v) \subseteq K_m(v)$ *for* $1 \leq m \leq n-1$.

The same holds for left eigenvectors, since $A^{\mathsf{T}} \in \mathcal{D}_n$.

Proof. (a) Since the column sums of A equal 1, $\sum_i (Av)_i = \sum_i v_i$, so $(\lambda - 1)\sum_i v_i = 0$.

(b) Let $|S| = m$ and $x_j = \sum_{i \in S} a_{ij}$. Then $\lambda \sum_{i \in S} v_i = \sum_{i \in S} (Av)_i = \sum_j x_j v_j$. The column sums give $0 \leq x_j \leq 1$ and the row sums give $\sum_j x_j = m$. The polytope $\{x \in [0,1]^n : \sum_j x_j = m\}$ (a hypersimplex) is the convex hull of its 0/1-points, i.e. of the indicator vectors of m-element sets, so $\lambda \sum_{i \in S} v_i \in K_m(v)$. As multiplication by λ is $\mathbb{R}$-linear, $\lambda K_m(v) \subseteq K_m(v)$. □

The case $m = 1$ uses only the row sums and is [12, Proposition 1]; the case $m = 2$ is [12, Corollary 2]. The general case also appears in the slides of [3, 2].

When the entries of v are convexly independent, the polygons $K_m(v)$ have a simple structure (this is also stated in [3, 2]). We only need it for $n = 6$ and $m = 2, 3$.

Lemma 2.2. *Let $v \in \mathbb{C}^6$ have convexly independent entries, labelled counterclockwise around their convex hull, and suppose $\sum_i v_i = 0$. Indices are taken modulo 6.*

(a) *$K_2(v)$ is a convex hexagon whose vertices, in counterclockwise order, are $s_j = v_j + v_{j+1}$, $j = 1, \dots, 6$. Moreover $s_1 + s_3 + s_5 = s_2 + s_4 + s_6 = 0$.*
(b) *$K_3(v)$ is a convex hexagon whose vertices are $t_j = v_j + v_{j+1} + v_{j+2}$, $j = 1, \dots, 6$, and $t_{j+3} = -t_j$.*

Proof. For a unit vector $u \in \mathbb{C}$ let $f_u(z) = \mathrm{dt}(u, z)$. Fix $m \in \{2, 3\}$ and write $t_j^{(m)} = v_j + \cdots + v_{j+m-1}$.

Every vertex of $K_m(v)$ is some $t_j^{(m)}$. A vertex of the polygon $K_m(v)$ is the unique maximizer of f_u over $K_m(v)$ for all u in an open arc of directions, so we may choose such a u for which f_u is injective on $\{v_1, \dots, v_6\}$. The maximizer is then the sum of the m entries with the largest values of f_u. A half-plane meets the boundary of the convex hexagon $K_1(v)$ in a connected arc, so these m entries are cyclically consecutive.

Every $t_j^{(m)}$ is a vertex, and they are distinct. Choose points p, q in the relative interiors of the edges $[v_{j-1}, v_j]$ and $[v_{j+m-1}, v_{j+m}]$ of $K_1(v)$. The line through p and q meets $\partial K_1(v)$ only at p, q, so it strictly separates $\{v_j, \dots, v_{j+m-1}\}$ from the remaining entries. If u is the normal of this line pointing towards the first set, then the subset $\{j, \dots, j + m - 1\}$ is the only m-subset maximizing f_u, and every other m-subset sum has a strictly smaller value of f_u. Hence $t_j^{(m)}$ is an exposed point of $K_m(v)$, different from all other subset sums. Consequently $K_m(v)$ has exactly the six vertices $t_1^{(m)}, \dots, t_6^{(m)}$.

The order of the vertices of $K_2(v)$. Let u be the normal of the line through v_j and v_{j+2} pointing towards v_{j+1}. Then $f_u(v_{j+1}) > f_u(v_j) = f_u(v_{j+2}) > f_u(v_i)$ for $i \in \{j + 3, j + 4, j + 5\}$, so the face of $K_2(v)$ in direction u is the segment $[s_j, s_{j+1}]$. Thus s_j and s_{j+1} are adjacent vertices for every j, and the boundary of $K_2(v)$ is the closed path $s_1 s_2 \cdots s_6$. It is traversed counterclockwise because

$$\mathrm{cr}(s_{j+1} - s_j,\ s_{j+2} - s_{j+1}) = \mathrm{cr}(v_{j+2} - v_j,\ v_{j+3} - v_{j+1}) > 0,$$

the cross product of the two diagonals of the counterclockwise convex quadrilateral $v_j v_{j+1} v_{j+2} v_{j+3}$.

Finally $s_1 + s_3 + s_5 = s_2 + s_4 + s_6 = \sum_i v_i = 0$, and $t_{j+3} = -t_j$ because $t_j + t_{j+3} = \sum_i v_i = 0$. □

2.2. **The Karpelevič region.** Let Θ_n be the set of eigenvalues of $n \times n$ (row) stochastic matrices. Taking direct sums with identity matrices shows $\Theta_m \subseteq \Theta_n$ for $m \le n$, and clearly $\omega_n \subseteq \Theta_n$. The region Θ_n was determined by Karpelevič [10]; see Ito [8] for a simpler statement and Munger, Nickerson and Paparella [14] for a modern account. We need two well-known facts.

Lemma 2.3. *For $n \ge 2$, Θ_n is star-shaped with respect to 0.*

Proof. Let B be stochastic with $Bv = \lambda v$, $v \neq 0$, and let $0 \le r \le 1$. If $\lambda = 1$, then r is an eigenvalue of the stochastic matrix $rI + (1 - r)\frac{1}{n}J$, where J is the all-ones matrix. If $\lambda \neq 1$, let $\pi \ge 0$ be a stationary distribution, $\pi^{\mathsf{T}} B = \pi^{\mathsf{T}}$, $\sum_i \pi_i = 1$. Then $\lambda \pi^{\mathsf{T}} v = \pi^{\mathsf{T}} B v = \pi^{\mathsf{T}} v$, so $\pi^{\mathsf{T}} v = 0$, and the stochastic matrix $B_r = rB + (1 - r)\mathbf{1}\pi^{\mathsf{T}}$ satisfies $B_r v = r\lambda v$. □

Theorem 2.4 (Dmitriev–Dynkin)**.** *If $\lambda \in \Theta_n$ and $0 \le \arg \lambda \le 2\pi/n$, then $\lambda \in \mathrm{conv}\{0, 1, e^{2\pi i/n}\}$. In particular, if $\lambda \in \Theta_6$ (for instance $\lambda \in \omega_6$) and $|\arg \lambda| \le \pi/3$, then $\lambda \in \Pi_6$.*

This goes back to Dmitriev and Dynkin [5, 6]; see the account in [11, p. 158]. It can also be read off from the Karpelevič theorem: inside the sector $0 \le \arg \lambda \le 2\pi/n$ the boundary of Θ_n is the segment $[1, e^{2\pi i/n}]$, see [14, Section 4.1]. Since Θ_n is compact, for $\lambda \in \Theta_n$ in this sector the point of Θ_n farthest from 0 on the ray through λ is a boundary point, hence lies on that segment; this gives the first statement. The second statement follows by applying the first one to λ or $\bar{\lambda}$, as Θ_6 and Π_6 are invariant under complex conjugation.

3. The sector $|\arg \lambda| \ge 2\pi/3$

Lemma 3.1. *Let C be a real $k \times k$ matrix whose columns have ℓ^1-norm at most 1, that is, $\sum_i |c_{ij}| \le 1$ for every j. Then every eigenvalue of C lies in Θ_{2k}.*

Proof. Write $C = C^+ - C^-$ with $C^\pm \geq 0$ entrywise and $c_{ij}^+ c_{ij}^- = 0$, and let

$$L = \begin{pmatrix} C^+ & C^- \\ C^- & C^+ \end{pmatrix}.$$

Then $L \geq 0$, every column sum of L equals $\sum_i |c_{ij}| \leq 1$, and if $Cx = \lambda x$ then $L\left(\begin{smallmatrix} x \\ -x \end{smallmatrix}\right) = \lambda\left(\begin{smallmatrix} x \\ -x \end{smallmatrix}\right)$. So λ is an eigenvalue of the nonnegative matrix $M = L^{\mathsf{T}}$, whose row sums are at most 1. After a simultaneous permutation of rows and columns, M is block upper triangular with irreducible diagonal blocks, and λ is an eigenvalue of one of these blocks, say M_0 of size $m \leq 2k$. The row sums of M_0 are at most 1, so its spectral radius ρ satisfies $\rho \leq 1$. If $\rho = 0$ then $\lambda = 0 \in \Theta_{2k}$. Otherwise, by the Perron–Frobenius theorem M_0 has a positive eigenvector p with $M_0 p = \rho p$, and $\rho^{-1} D^{-1} M_0 D$ with $D = \operatorname{diag}(p)$ is stochastic and has the eigenvalue λ/ρ. Hence $\lambda/\rho \in \Theta_m \subseteq \Theta_{2k}$, and $\lambda = \rho \cdot (\lambda/\rho) \in \Theta_{2k}$ by Lemma 2.3. □

Proposition 3.2. *Let $A \in \mathcal{D}_6$, $Av = \lambda v$ with $\lambda \notin \mathbb{R}$, and suppose the entries of v are convexly independent. Then λ is an eigenvalue of a real 3×3 matrix whose columns have ℓ^1-norm at most 1. Consequently, if $|\arg \lambda| \geq 2\pi/3$, then $\lambda \in \Pi_6$.*

Proof. After a simultaneous permutation of the rows and columns of A we may assume that $v_1, \ldots, v_6$ are labelled counterclockwise. By Lemma 2.1, $\sum_i v_i = 0$ and $\lambda K_3(v) \subseteq K_3(v)$, and by Lemma 2.2(b), $K_3(v)$ is the hexagon with vertices $\pm w_1, \pm w_2, \pm w_3$, where $w_j = t_j$. Hence

$$K_3(v) = \Big\{ \sum_{j=1}^{3} c_j w_j : \ c \in \mathbb{R}^3, \ \sum_j |c_j| \leq 1 \Big\},$$

the image of the ℓ^1 unit ball under the $\mathbb{R}$-linear map $W : \mathbb{R}^3 \to \mathbb{C} \cong \mathbb{R}^2$, $We_j = w_j$, which is onto because $K_3(v)$ has nonempty interior. Since $\lambda w_j \in K_3(v)$, there is a real 3×3 matrix $C = (c_{ij})$ with $\sum_i |c_{ij}| \leq 1$ and $\lambda w_j = \sum_i c_{ij} w_i$ for each j; in matrix form $M_\lambda W = WC$, where $M_\lambda = \left(\begin{smallmatrix} \operatorname{Re}\lambda & -\operatorname{Im}\lambda \\ \operatorname{Im}\lambda & \operatorname{Re}\lambda \end{smallmatrix}\right)$ represents multiplication by λ. The real matrix M_λ has the eigenvalue λ; let $y \in \mathbb{C}^{1\times 2}$ be a left eigenvector, $yM_\lambda = \lambda y$. Then $(yW)C = yM_\lambda W = \lambda(yW)$, and $yW \neq 0$ because W is onto. So λ is an eigenvalue of C.

For the second statement, note that $-C$ and C satisfy the same hypothesis and that C is real, so $\bar\lambda$, $-\lambda$ and $-\bar\lambda$ are eigenvalues of such matrices as well. By Lemma 3.1, $-\bar\lambda \in \Theta_6$. If $2\pi/3 \leq |\arg\lambda| \leq \pi$, then $|\arg(-\bar\lambda)| \leq \pi/3$, and Theorem 2.4 gives $-\bar\lambda \in \Pi_6$. Since Π_6 is invariant under $z \mapsto -\bar z$, we get $\lambda \in \Pi_6$. □

Remark 3.3. Proposition 3.2 alone does not suffice in the middle sector. For example, the matrix

$$C = \begin{pmatrix} 0 & 0 & 1/4 \\ 1 & 0 & -3/4 \\ 0 & 1 & 0 \end{pmatrix}$$

has columns of ℓ^1-norm 1 and characteristic polynomial $\lambda^3 + \frac{3}{4}\lambda - \frac{1}{4}$, whose non-real roots $\lambda \approx -0.1490 \pm 0.9037i$ (with $\arg\lambda \approx \pm 99.4^\circ$) lie outside PM_6. Likewise, using only $\lambda K_1(v) \subseteq K_1(v)$, i.e. $\lambda \in \Theta_6$, is not enough, since $\Theta_6 \not\subseteq \mathrm{PM}_6$.

4. The middle sector and Proposition P_{90}

Definition 4.1. A hexagon $H = \operatorname{conv}\{s_1, \ldots, s_6\} \subset \mathbb{C}$ is *admissible* if $s_1, \ldots, s_6$ are its six vertices in counterclockwise order (so H is strictly convex) and

$$s_1 + s_3 + s_5 = s_2 + s_4 + s_6 = 0.$$

We write $T_1 = \operatorname{conv}\{s_1, s_3, s_5\}$ and $T_2 = \operatorname{conv}\{s_2, s_4, s_6\}$; both triangles have centroid 0. A point $x \in \mathbb{C} \setminus \{0\}$ is a *witness* for H if $\operatorname{Im}(s_j/x) \leq \frac{\sqrt{3}}{2}$ for $j = 1, \ldots, 6$.

By Lemma 2.2(a), if $v \in \mathbb{C}^6$ has convexly independent entries labelled counterclockwise and $\sum_i v_i = 0$, then $K_2(v)$ is admissible with $s_j = v_j + v_{j+1}$.

Lemma 4.2. *Let $H \subset \mathbb{C}$ be compact and convex, and let $x \in H \setminus \{0\}$ satisfy $\operatorname{Im}(z/x) \le \frac{\sqrt{3}}{2}$ for all $z \in H$. Then $\operatorname{Im}\lambda \le \frac{\sqrt{3}}{2}$ for every $\lambda \in S(H)$. If H is a polygon, it suffices to check the condition on x at the vertices of H.*

Proof. If $\lambda H \subseteq H$ then $\lambda x \in H$, so $\operatorname{Im}\lambda = \operatorname{Im}(\lambda x/x) \le \frac{\sqrt{3}}{2}$. The last statement holds because $z \mapsto \operatorname{Im}(z/x)$ is $\mathbb{R}$-linear. □

Proposition 4.3 (P_{90})**.** *Every admissible hexagon H has a witness $x \in \partial H$; that is, there is $x \in \partial H$ with*

$$\operatorname{Im}\frac{s_j}{x} \le \frac{\sqrt{3}}{2} \qquad (j = 1, \dots, 6).$$

Since $\operatorname{Im}(z/x) = \operatorname{Re}\big(z\,\overline{ix}\big)/|x|^2$, the condition on x is equivalent to $h_H(\arg x + \pi/2) \le \frac{\sqrt{3}}{2}|x|$: it concerns the support of H in the direction of x rotated by 90°, whence the name. Sections 5 and 6 contain the proof. We first deduce the main theorem.

Proof of Theorem 1.1. Replacing A by A^{T} if necessary, we may assume that $Av = \lambda v$ for a vector v with convexly independent entries. Real eigenvalues lie in $[-1, 1] \subseteq \Pi_6$, so let $\lambda \notin \mathbb{R}$. Since $A\bar{v} = \bar{\lambda}\bar{v}$, the entries of $\bar{v}$ are convexly independent and $\Pi_6 = \overline{\Pi_6}$, we may assume $\operatorname{Im}\lambda > 0$. After a simultaneous permutation of rows and columns of A, the entries $v_1, \dots, v_6$ are labelled counterclockwise. By Lemma 2.1, $\sum_i v_i = 0$ and $\lambda K_m(v) \subseteq K_m(v)$ for all m.

- If $0 < \arg\lambda \le \pi/3$, then $\lambda \in \Pi_6$ by Theorem 2.4.
- If $2\pi/3 \le \arg\lambda < \pi$, then $\lambda \in \Pi_6$ by Proposition 3.2.
- If $\pi/3 < \arg\lambda < 2\pi/3$, then $H = K_2(v)$ is admissible and $\lambda \in S(H)$. By Proposition 4.3 and Lemma 4.2, $\operatorname{Im}\lambda \le \frac{\sqrt{3}}{2}$. In the sector $\pi/3 \le \arg\lambda \le 2\pi/3$ the hexagon Π_6 coincides with the triangle $\operatorname{conv}\{0, \zeta, \omega\}$, i.e. with the set of points of the sector with $\operatorname{Im}\lambda \le \frac{\sqrt{3}}{2}$. Hence $\lambda \in \Pi_6$. □

Remark 4.4 (Sharpness of P_{90}). Suppose $s_{k+2} = \omega s_k$ for all k, i.e. H is invariant under the rotation by $2\pi/3$. Then $\omega \in S(H)$ and $\operatorname{Im}\omega = \frac{\sqrt{3}}{2}$. The proof of Lemma 4.2 shows that if some $x \in H \setminus \{0\}$ satisfied $\operatorname{Im}(z/x) < \frac{\sqrt{3}}{2}$ for all $z \in H$, then every $\lambda \in S(H)$ would satisfy $\operatorname{Im}\lambda < \frac{\sqrt{3}}{2}$. Hence no such x exists: the inequality of Proposition 4.3 has zero margin for every such H. Up to rotation and dilation these hexagons form a two-parameter family ($s_1 = 1$, $s_2 = re^{i\alpha}$), which contains the regular hexagon and degenerates to the "doubled triangles" $(1, 1, \omega, \omega, \omega^2, \omega^2)$ and $(1, \omega, \omega, \omega^2, \omega^2, 1)$. Consequently, any proof of Proposition 4.3 by interval arithmetic must avoid estimating the margin of P_{90} directly near this family; this is the purpose of the lemmas in Section 5. Note also that the eigenvalue ω is attained: if A is the permutation matrix of two disjoint 3-cycles, the ω-eigenspace of A contains convexly independent vectors, and their hexagons $K_2(v)$ belong to this family.

5. Proof of Proposition P_{90}: three potential lemmas

Throughout this section H is an admissible hexagon. For $k \in \mathbb{Z}$ and $d \in \{1, \dots, 5\}$ we denote by (I_k^d) the inequality

$$(I_k^d) \qquad \operatorname{Im}\frac{s_{k+d}}{s_k} \le \frac{\sqrt{3}}{2}.$$

Thus the vertex s_k is a witness if and only if (I_k^d) holds for $d = 1, \dots, 5$.

5.1. **Elementary facts.** The triangles T_1, T_2 are nondegenerate (their vertices are vertices of the strictly convex hexagon H) and have centroid 0, so $0 \in \operatorname{int} T_1 \cap \operatorname{int} T_2$. Since 0 is an interior point of H and $s_1, \dots, s_6$ are the vertices of H in counterclockwise order, we can write $s_{k+1}/s_k = |s_{k+1}/s_k| e^{ia_k}$ with

$$a_k \in (0, \pi), \qquad a_{k+6} = a_k, \qquad \sum_{k=1}^{6} a_k = 2\pi.$$

Moreover $a_k + a_{k+1} \in (0, \pi)$, as this is the angle at 0 subtended by the side $[s_k, s_{k+2}]$ of T_1 or T_2, and 0 is interior to that triangle. Fix θ_1 with $s_1 = |s_1| e^{i\theta_1}$ and define $\theta_{k+1} = \theta_k + a_k$ for all $k \in \mathbb{Z}$. Then $s_k = |s_k| e^{i\theta_k}$ and $\theta_{k+6} = \theta_k + 2\pi$.

Lemma 5.1. *For every $k \in \mathbb{Z}$:*

(a) $\operatorname{Im}(s_{k+4}/s_k) < 0$ *and* $\operatorname{Im}(s_{k+5}/s_k) < 0$*; in particular* (I_k^4) *and* (I_k^5) *hold. Hence s_k is a witness if and only if* (I_k^1)*,* (I_k^2) *and* (I_k^3) *hold.*

(b) *Let $d \in \{1, 2, 3\}$, $\varphi = \theta_{k+d} - \theta_k \in (0, 2\pi)$ and $L = \log|s_{k+d}/s_k|$. If $\varphi \geq \pi$, then* (I_k^d) *holds. If $\varphi < \pi$, then* (I_k^d) *is equivalent to* $L + \log\sin\varphi \leq \log\frac{\sqrt{3}}{2}$.

(c) *For $0 < \varphi < \pi$,*

$$\log\sin\varphi \leq \log\frac{\sqrt{3}}{2} + \frac{\varphi - \pi/3}{\sqrt{3}} \qquad \text{and} \qquad \log\sin\varphi \leq \log\frac{\sqrt{3}}{2} - \frac{\varphi - 2\pi/3}{\sqrt{3}}. \tag{1}$$

(d) $\operatorname{Im}\dfrac{s_{k+2}}{s_k} = \left|\dfrac{s_{k-2}}{s_k}\right| \sin(\theta_k - \theta_{k-2})$*, and* $\theta_k - \theta_{k-2} \in (0, \pi)$.

Proof. (a) We have $\arg(s_{k+5}/s_k) = \theta_{k-1} - \theta_k = -a_{k-1} \in (-\pi, 0)$ and $\arg(s_{k+4}/s_k) = \theta_{k-2} - \theta_k = -(a_{k-2} + a_{k-1}) \in (-\pi, 0)$.

(b) φ is a sum of $d \leq 3$ of the angles a_i, hence lies in $(0, 2\pi)$, and $\operatorname{Im}(s_{k+d}/s_k) = e^L \sin\varphi$. If $\varphi \geq \pi$ this is ≤ 0; otherwise take logarithms.

(c) $\log\sin$ is concave on $(0, \pi)$, and the right-hand sides are its tangent lines at $\pi/3$ and $2\pi/3$, where its derivative $\cot\varphi$ equals $\pm 1/\sqrt{3}$.

(d) If $a + b + c = 0$ then $\operatorname{cr}(a, b) = \operatorname{cr}(b, c) = \operatorname{cr}(c, a)$; for instance $\operatorname{cr}(b, c) = \operatorname{cr}(b, -a - b) = -\operatorname{cr}(b, a) = \operatorname{cr}(a, b)$. Applying this to the triangle $(a, b, c) = (s_{k-2}, s_k, s_{k+2})$, whose vertices sum to 0, gives

$$\operatorname{Im}\frac{s_{k+2}}{s_k} = \frac{\operatorname{cr}(s_k, s_{k+2})}{|s_k|^2} = \frac{\operatorname{cr}(s_{k-2}, s_k)}{|s_k|^2} = \frac{|s_{k-2}|\,|s_k| \sin(\theta_k - \theta_{k-2})}{|s_k|^2},$$

and $\theta_k - \theta_{k-2} = a_{k-2} + a_{k-1} \in (0, \pi)$. □

5.2. **The largest vertex of a triangle.**

Lemma 5.2. *If $|s_k| \geq \max(|s_{k+2}|, |s_{k+4}|)$, then $|\operatorname{Im}(s_{k+2}/s_k)| \leq \frac{\sqrt{3}}{2}$ and $|\operatorname{Im}(s_{k+4}/s_k)| \leq \frac{\sqrt{3}}{2}$. In particular* (I_k^2) *holds.*

Proof. Let $u = s_{k+2}/s_k$. Then $s_{k+4}/s_k = -1 - u$, and the hypothesis gives $|u| \leq 1$ and $|1 + u| \leq 1$. The intersection of the closed unit discs centred at 0 and -1 is a lens with corners $-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$, so $|\operatorname{Im} u| \leq \frac{\sqrt{3}}{2}$; the same applies to $-1 - u$. □

5.3. **Potentials.** For $k \in \mathbb{Z}$ define

$$\begin{aligned} \Psi_k^A &= \tfrac{\sqrt{3}}{2} \log|s_k| + \tfrac{1}{2}\theta_k - \tfrac{(k-1)\pi}{6}, \\ \Psi_k^B &= \tfrac{\sqrt{3}}{2} \log|s_k| - \tfrac{1}{2}\theta_k + c_k^B, \\ \Psi_k^C &= \tfrac{\sqrt{3}}{2} \log|s_k| - \tfrac{1}{2}\theta_k + c_k^C, \end{aligned}$$

where

$$(c_1^B, \dots, c_6^B) = \big(0, 0, \tfrac{\pi}{3}, \tfrac{\pi}{3}, \tfrac{2\pi}{3}, \tfrac{2\pi}{3}\big), \qquad (c_1^C, \dots, c_6^C) = \big(0, \tfrac{\pi}{3}, \tfrac{\pi}{3}, \tfrac{2\pi}{3}, \tfrac{2\pi}{3}, \pi\big),$$

extended to all $k \in \mathbb{Z}$ by $c_{k+6} = c_k + \pi$. Since $\theta_{k+6} = \theta_k + 2\pi$, all three sequences are 6-periodic in k, so each of them attains its maximum. The following table lists $(c_{k+d} - c_k)/(\pi/3)$ for $d = 1, 2, 3$; it depends only on the parity of k.

	$d=1$	$d=2$	$d=3$
c^B, k odd	0	1	1
c^B, k even	1	1	2
c^C, k odd	1	1	2
c^C, k even	0	1	1

Lemma 5.3 (Lemma A)**.** *If $\Psi_k^A = \max_j \Psi_j^A$, then (I_k^1) and (I_k^2) hold.*

Proof. (I_k^1): here $\varphi = a_k \in (0, \pi)$. From $\Psi_{k+1}^A \le \Psi_k^A$ we get $\frac{\sqrt{3}}{2}L + \frac{1}{2}a_k - \frac{\pi}{6} \le 0$, i.e. $L \le -(\varphi - \pi/3)/\sqrt{3}$. Adding the first inequality of (1) gives $L + \log\sin\varphi \le \log\frac{\sqrt{3}}{2}$, which is (I_k^1) by Lemma 5.1(b).

(I_k^2): let $\psi = \theta_k - \theta_{k-2} \in (0, \pi)$. From $\Psi_{k-2}^A \le \Psi_k^A$ we get $\frac{\sqrt{3}}{2}\log|s_{k-2}/s_k| \le \frac{1}{2}\psi - \frac{\pi}{3}$, i.e. $\log|s_{k-2}/s_k| \le (\psi - 2\pi/3)/\sqrt{3}$. Adding the second inequality of (1) with $\varphi = \psi$ gives $\log|s_{k-2}/s_k| + \log\sin\psi \le \log\frac{\sqrt{3}}{2}$, which is (I_k^2) by Lemma 5.1(d). □

Lemma 5.4 (Lemmas B and C)**.** *Let $c \in \{c^B, c^C\}$ and $\Psi = \Psi^B$ or Ψ^C accordingly. If $\Psi_k = \max_j \Psi_j$ and $d \in \{1, 2, 3\}$ satisfies $c_{k+d} - c_k \ge \pi/3$, then (I_k^d) holds. Consequently, by the table above and Lemma 5.1(a):*

(i) *if k maximizes Ψ^B, then (I_k^2) and (I_k^3) hold, and s_k is a witness when k is even;*
(ii) *if k maximizes Ψ^C, then (I_k^2) and (I_k^3) hold, and s_k is a witness when k is odd.*

Proof. Let $\varphi = \theta_{k+d} - \theta_k$ and $L = \log|s_{k+d}/s_k|$. If $\varphi \ge \pi$ there is nothing to prove. Otherwise $\Psi_{k+d} \le \Psi_k$ gives $\frac{\sqrt{3}}{2}L \le \frac{1}{2}\varphi - (c_{k+d} - c_k) \le \frac{1}{2}\varphi - \frac{\pi}{3}$, i.e. $L \le (\varphi - 2\pi/3)/\sqrt{3}$, and the second inequality of (1) yields $L + \log\sin\varphi \le \log\frac{\sqrt{3}}{2}$. □

Remark 5.5 (How the lemmas are used). Each lemma reduces "s_k is a witness" to a single remaining inequality: (I_k^3) for Lemma 5.3, and (I_k^1) for Lemma 5.4 when k has the "wrong" parity. The point is that these remaining inequalities have positive margin where P_{90} itself is tight. For the regular hexagon $s_k = \zeta^{k-1}$, the inequalities (I_k^1) and (I_k^2) hold with equality while (I_k^3) reads $0 \le \frac{\sqrt{3}}{2}$; near the doubled triangle $(1, 1, \omega, \omega, \omega^2, \omega^2)$ the remaining inequality of Lemma 5.4(i) compares two vertices of the same cluster and has margin close to $\frac{\sqrt{3}}{2}$, and similarly for Lemma 5.4(ii) near $(1, \omega, \omega, \omega^2, \omega^2, 1)$. Along the whole family of Remark 4.4, numerical experiments indicate that at least one of the certificates (A), (B), (C), (L_1), (L_2) of Section 6 holds with a margin of at least about 0.1 (for the quantities q_j of Section 6, normalized by $\max_k|s_k| = 1$). The proof does not depend on this observation: the computation in Section 6 checks directly that on every box one of the certificates holds.

Remark 5.6 (Vertices do not suffice). No rule that selects a vertex can prove P_{90} on its own: there are admissible hexagons without a vertex witness. For example, for $s_1 = 1$, $s_2 = 0.08 + 0.89i$, $s_3 = -0.74 + 0.54i$, $s_4 = -0.53 - 0.64i$, $s_5 = -1 - s_3$, $s_6 = -s_2 - s_4$, an exact rational computation shows that the hexagon is admissible, that each vertex s_k violates some (I_k^d), and that the boundary point $x = \frac{21}{64}s_3 + \frac{43}{64}s_4$ is a witness. Such hexagons are rare (a few per million random samples in our experiments), but they force us to allow witnesses on edges.

6. Proof of Proposition P_{90}: the computation

6.1. **Normalization.** Cyclic relabelling $s_k \mapsto s_{k+1}$ maps admissible hexagons to admissible hexagons (it exchanges T_1 and T_2), and so does $z \mapsto z/c$ for $c \in \mathbb{C}\setminus\{0\}$. Both operations preserve the conclusion

of Proposition 4.3, since $\operatorname{Im}\big((s_j/c)/(x/c)\big) = \operatorname{Im}(s_j/x)$. Hence we may assume that

$$s_1 = 1 \quad \text{and} \quad |s_k| \le 1 \quad (k = 2, \dots, 6).$$

Such a hexagon is determined by $p = (p_1, \dots, p_6) = (\operatorname{Re} s_2, \operatorname{Im} s_2, \operatorname{Re} s_3, \operatorname{Im} s_3, \operatorname{Re} s_4, \operatorname{Im} s_4) \in \mathbb{R}^6$, through

$$s_1 = 1, \quad s_2 = p_1 + ip_2, \quad s_3 = p_3 + ip_4, \quad s_4 = p_5 + ip_6, \quad s_5 = -1 - s_3, \quad s_6 = -s_2 - s_4.$$

Let $\Omega \subset \mathbb{R}^6$ be the set of p for which $(s_1, \dots, s_6)$ is admissible and $\max_k |s_k| = 1$. Proposition 4.3 is equivalent to the statement that for every $p \in \Omega$ the hexagon H has a witness on ∂H.

Lemma 6.1. $\Omega \subseteq B_0 := [-1, 1] \times [0, 1] \times [-1, 0] \times [0, \frac{7}{8}] \times [-1, 1] \times [-1, 1]$.

Proof. All coordinates are bounded by 1 in absolute value because $|s_k| \le 1$. With $\theta_1 = 0$ we have $\theta_2 = a_1 \in (0, \pi)$ and $\theta_3 = a_1 + a_2 \in (0, \pi)$, so $\operatorname{Im} s_2 > 0$ and $\operatorname{Im} s_3 > 0$. Since $|s_3| \le 1$ and $|1 + s_3| = |s_5| \le 1$, we get $2 \operatorname{Re} s_3 \le -|s_3|^2 \le 0$, and s_3 lies in the lens of Lemma 5.2, so $\operatorname{Im} s_3 \le \frac{\sqrt{3}}{2} < \frac{7}{8}$. □

6.2. **Quantities and enclosures.** All quantities that we bound are either quadratic polynomials in p or are obtained from such polynomials by log and arctan. The quadratic polynomials are

- $|s_k|^2$, $\operatorname{cr}(s_k, s_{k+1})$ and $\operatorname{dt}(s_k, s_{k+1})$;
- the turns $\tau_k = \operatorname{cr}(s_{k+1} - s_k,\ s_{k+2} - s_{k+1})$ and the 24 edge–vertex products $\epsilon_{k,j} = \operatorname{cr}(s_{k+1} - s_k,\ s_j - s_k)$, $j \notin \{k, k+1\}$;
- for a point z that is an affine function of p (a vertex, or a point $(1-t)s_k + ts_{k+1}$ of an edge with fixed t),

$$q_j(z) = \tfrac{\sqrt{3}}{2}|z|^2 - \operatorname{cr}(z, s_j),$$

so that $q_j(z) \ge 0$ if and only if $\operatorname{Im}(s_j/z) \le \frac{\sqrt{3}}{2}$ (for $z \ne 0$).

Each such polynomial is $P(p) = y^{\mathsf{T}} G y$ with $y = (p, 1) \in \mathbb{R}^7$ and a symmetric 7×7 matrix $G = G_0 + \sqrt{3}\, G_1$, where G_0, G_1 have dyadic rational entries. On a box with centre m and half-widths h we use the centred form: writing $p = m + \delta$ with $|\delta_i| \le h_i$ and $Q = (G_{il})_{i,l \le 6}$,

$$P(m + \delta) = P(m) + 2 \sum_{i \le 6} (G y_m)_i\, \delta_i + \delta^{\mathsf{T}} Q \delta, \qquad y_m = (m, 1),$$

where $|2 \sum_i (G y_m)_i \delta_i| \le 2 \sum_i |(G y_m)_i| h_i$ and

$$\sum_i \min(Q_{ii}, 0) h_i^2 - \sum_{i \ne l} |Q_{il}| h_i h_l \ \le\ \delta^{\mathsf{T}} Q \delta \ \le\ \sum_i \max(Q_{ii}, 0) h_i^2 + \sum_{i \ne l} |Q_{il}| h_i h_l.$$

The angle a_k is the argument of $\operatorname{dt}(s_k, s_{k+1}) + i \operatorname{cr}(s_k, s_{k+1})$, which lies in $(0, \pi)$ on Ω. On a box where the enclosure of $\operatorname{cr}(s_k, s_{k+1})$ is positive we enclose $a_k = \frac{\pi}{2} - \arctan(\operatorname{dt} / \operatorname{cr})$; where instead the enclosure of $\operatorname{dt}(s_k, s_{k+1})$ is positive we enclose $a_k = \arctan(\operatorname{cr} / \operatorname{dt})$ and intersect with $[0, \infty)$. Then $\theta_k = a_1 + \cdots + a_{k-1}$ (with $\theta_1 = 0$), and a difference $\theta_k - \theta_j$ is enclosed as a sum of the enclosures of the a_i between j and k. Finally $\log|s_k| = \frac{1}{2} \log|s_k|^2$, where we use the lower bound $-\infty$ whenever the enclosure of $|s_k|^2$ contains 0. The enclosures of a_k, θ_k and Ψ_k^X are valid at every point of $B \cap \Omega$, which is all we need.

6.3. **Discarded and certified boxes.** A box $B \subseteq B_0$ is *discarded* if, for one of the following necessary conditions for $p \in \Omega$, the enclosure shows that the condition fails at every point of B:

(N1) $|s_k|^2 \le 1$ for $k = 2, \dots, 6$;
(N2) $\tau_k \ge 0$ for all k (left turns);
(N3) $\operatorname{cr}(s_k, s_{k+1}) \ge 0$ for all k;
(N4) $\epsilon_{k,j} \ge 0$ for all k and $j \notin \{k, k+1\}$ (every vertex lies to the left of every edge);

(N5) $\sum_{k=1}^{6} a_k = 2\pi$; the box is discarded when a lower bound of $\sum_k a_k$ exceeds 2π. (In the final cover this condition is never needed.)

A box B is *certified* if one of the following holds, where every inequality is verified with the enclosures above on the whole of B:

(W_v) *vertex witness*: for some k, $q_j(s_k) \geq 0$ for all $j \neq k$;
(W_e) *edge witness*: for some k and some $t \in 2^{-12}\mathbb{Z} \cap [0,1]$, the point $z = (1-t)s_k + ts_{k+1}$ satisfies $q_j(z) \geq 0$ for all j;
(A) for every k in the candidate set $\mathcal{K}^A(B)$, $q_{k+3}(s_k) \geq 0$;
(B) for every odd k in the candidate set $\mathcal{K}^B(B)$, $q_{k+1}(s_k) \geq 0$;
(C) for every even k in the candidate set $\mathcal{K}^C(B)$, $q_{k+1}(s_k) \geq 0$;
(L_1) for every $k \in \{1,3,5\}$ for which there is no $k' \in \{1,3,5\}$ with $\sup_B |s_k|^2 < \inf_B |s_{k'}|^2$ (sup and inf taken from the enclosures), we have $q_j(s_k) \geq 0$ for $j \in \{2,4,6\}$;
(L_2) the same with $\{1,3,5\}$ and $\{2,4,6\}$ exchanged.

Here, for $X \in \{A,B,C\}$, the candidate set $\mathcal{K}^X(B)$ consists of those $k \in \{1,\dots,6\}$ for which there is no j with an upper bound for $\Psi_k^X - \Psi_j^X$ on B that is negative. Thus $\mathcal{K}^X(B)$ contains every maximizer of Ψ^X, at every point of $B \cap \Omega$.

Lemma 6.2. *If B is discarded, then $B \cap \Omega = \emptyset$. If B is certified, then every hexagon with parameters in $B \cap \Omega$ has a witness on its boundary.*

Proof. For $p \in \Omega$ the conditions (N1)–(N5) hold: (N2) and (N4) express strict convexity and the counterclockwise order, (N3) and (N5) hold because $0 \in \operatorname{int} H$ and the $a_k \in (0,\pi)$ sum to 2π. This proves the first claim. Now let $p \in B \cap \Omega$; all enclosures above are valid at p. In cases (W_v) and (W_e) the point s_k, respectively z, lies on ∂H, hence $z \neq 0$ because $0 \in \operatorname{int} H$, and it is a witness. In case (A), let k maximize Ψ^A at p; then $k \in \mathcal{K}^A(B)$, so (I_k^3) holds, (I_k^1) and (I_k^2) hold by Lemma 5.3, and s_k is a witness by Lemma 5.1(a). In case (B), let k maximize Ψ^B at p. If k is even, s_k is a witness by Lemma 5.4(i). If k is odd, then (I_k^2) and (I_k^3) hold by Lemma 5.4(i) and (I_k^1) holds by (B), so again s_k is a witness. Case (C) is symmetric. In case (L_1), let $k \in \{1,3,5\}$ be such that $|s_k|$ is maximal in T_1 at p; then k is one of the indices checked in (L_1), so (I_k^1), (I_k^3), (I_k^5) hold, while (I_k^2) and (I_k^4) hold by Lemma 5.2; so s_k is a witness. Case (L_2) is the same. □

6.4. **The computation.** A floating-point branch-and-bound search starts from B_0 and bisects every box that is neither discarded nor certified along its (first) widest coordinate; for (W_e) it tries the best edge point at the centre of the box, with t rounded to a multiple of 2^{-12}. The search terminates after a few seconds with 53,093 boxes, distributed as follows.

outcome	boxes
discarded by (N1) / (N2) / (N3) / (N4)	6,635 / 5,943 / 96 / 4,846
(W_v) vertex witness	19,771
(W_e) edge witness	9,988
(A)	3,644
(B)	936
(C)	1,191
(L_1) / (L_2)	28 / 15
total	53,093

Without the 24 conditions in (N4), the search keeps subdividing boxes that contain no convex hexagon at all; with them it terminates quickly.

The output of the search is not trusted. A separate program re-checks every box using Arb ball arithmetic [9] through python-flint (version 0.9.0) at 96 bits of precision. It reads only the boxes (centres and half-widths are exact binary floating-point numbers), the certificate types and the

data k, t, recomputes all enclosures with exact coefficients (with $\sqrt{3}$ and π represented by balls), and accepts an inequality only if it holds for the whole ball. The coefficient matrices G_0, G_1 are computed in floating point and compared entry by entry with the same matrices computed in exact rational arithmetic before they are used. All 53,093 boxes pass. The program also checks that the boxes tile B_0: it replays the bisection rule starting from B_0 and verifies that every branch ends in one of the listed boxes (and, as a consistency check, that the total volume equals that of B_0 in exact rational arithmetic). Before the run it tests the enclosures on random boxes and points. The verification takes about half a minute on two cores.

By Lemmas 6.1 and 6.2, every $p \in \Omega$ lies in some certified box, which proves Proposition 4.3 and hence Theorem 1.1.

6.5. **Independent checks.** To guard against errors in the implementation we also ran the following checks.

- *An independent verifier*, written without reusing any of the code above. It uses first-order affine arithmetic with Arb balls (80 bits), its own enclosures of the angles, its own implementation of the potentials and candidate sets, and accepts a box if it is discarded, if its recorded certificate holds, or if one of the certificate types other than (W_e) holds; otherwise it bisects the box, up to 12 further times. All boxes are confirmed (0 unresolved), and the tiling is confirmed independently (about 170 seconds).
- *Mutation tests* (script and log included): every corruption of the certificate data that we tried makes the verification fail. This includes changing the witness vertex of 300 boxes, moving the edge parameter t of 300 boxes by $\frac{1}{2}$, relabelling boxes with another certificate type, changing the recorded reason for discarding a box, declaring certified boxes discarded, deleting or duplicating a box, and replacing a box by one of its halves. (Moving t by 2^{-12} is not detected, as expected: the shifted point is usually still a witness.) Changing the constants of Ψ^A (flipping their sign, setting them to 0, or shifting their indices) also makes the verification fail, but only through 75–85 of the 3,644 boxes of type (A). Changing the constants c^B, c^C in the same ways is *not* detected, because almost all boxes of types (B) and (C) satisfy the checked inequality for all candidate indices. The correctness of these constants therefore rests on the proofs of Lemmas 5.3 and 5.4 and on the table in Section 5, which are also tested numerically:
- *Numerical tests of the lemmas*: Lemmas 5.1–5.4 were tested on 240,000 random admissible hexagons, including samples concentrated near the regular hexagon, the two doubled triangles and hexagons with nearly collinear vertices, with random cyclic relabelling and rotation; there were no violations beyond a floating-point tolerance of 10^{-12}.
- *Consistency of the cover*: for 20,000 random normalized admissible hexagons, the box containing the point and its certificate were checked pointwise.
- *End-to-end*: among 200,000 random matrices in $\mathcal{D}_6$ (convex combinations of 2 to 8 random permutation matrices) and their transposes, 134,586 non-real eigenvalues had an eigenvector with convexly independent entries; all of them lie in Π_6. For 1,508 random convexly independent vectors with zero sum, the vertex set of $K_2(v)$ was confirmed to be $\{s_1, \dots, s_6\}$ as in Lemma 2.2(a).

The scripts and the certificate file are available at

`https://github.com/jackelinemontalvo221-cyber/lpk-conjecture-n6`

and the verification is reproduced by running `python3 verify_arb.py cover.npz` in the directory `proof`.

7. Concluding remarks

What the proof uses. The inner sector uses only that A is stochastic. The outer sector uses only $\lambda K_3(v) \subseteq K_3(v)$, and the middle sector only $\lambda K_2(v) \subseteq K_2(v)$. In particular, Proposition 4.3

shows that for convexly independent $v \in \mathbb{C}^6$ with $\sum_i v_i = 0$ the single condition $\lambda K_2(v) \subseteq K_2(v)$ already forces $\operatorname{Im}\lambda \le \frac{\sqrt{3}}{2}$. This is the exact analogue, one dimension up, of the situation for $n = 4$ in [12], where $K_2(v)$ is a parallelogram made of two segments with midpoint 0, and the condition $\lambda K_2(v) \subseteq K_2(v)$ forces $\lambda \in \Pi_4$.

Larger n. The method does not extend directly. In numerical experiments with $n = 8$ we found convexly independent $v \in \mathbb{C}^8$ with $\sum_i v_i = 0$ and λ outside Π_8 (by about 1.9% in the radial direction, at $\arg\lambda \approx 70^\circ$) such that $\lambda K_2(v) \subseteq K_2(v)$; so for $n = 8$ the condition $\lambda K_2(v) \subseteq K_2(v)$ alone does not imply $\lambda \in \Pi_8$. We have not verified this observation rigorously.

The Perfect–Mirsky conjecture for $n = 6$. By Corollary 1.2, a counterexample $\lambda \in \omega_6 \setminus \mathrm{PM}_6$ would require that no left or right eigenvector for λ has convexly independent entries, i.e. that the convex hulls of the entries of these eigenvectors have at most five vertices. This case remains open.

Use of AI tools

This work was carried out with substantial assistance from the AI system Claude (Anthropic). In particular, the reduction to Proposition 4.3, the potential lemmas of Section 5 and the programs of Section 6 were developed in interactive sessions with Claude, and the independent verifier was written by a separate instance of Claude that was instructed to search for errors. The author is responsible for the content of this note.

References

[1] L. Bouthat, J. Mashreghi and F. Morneau-Guérin, *The doubly stochastic inverse eigenvalue problem*, poster, Heidelberg Laureate Forum, September 2025. https://r-libre.teluq.ca/4010

[2] L. Bouthat, J. Mashreghi and F. Morneau-Guérin, *New results on the doubly stochastic inverse eigenvalue problem*, talk at ILAS 2026, Virginia Tech, May 21, 2026. Slides: https://r-libre.teluq.ca/4184/1/Beamer_ILAS.pdf

[3] L. Bouthat, J. Mashreghi and F. Morneau-Guérin, *Toward a solution to a conjecture on doubly stochastic eigenvalues*, slides of a talk by L. Bouthat at the Analysis Seminar, University of Delaware, May 15, 2026 (announced as "A complete solution to a conjecture on doubly stochastic eigenvalues"). https://r-libre.teluq.ca/4086/1/Beamer_Delaware.pdf

[4] L. Bouthat, J. Mashreghi and F. Morneau-Guérin, *A complete solution to a conjecture on doubly stochastic eigenvalues*, another version of the slides of the talk [3]; file attached to the r-libre record deposited on May 25, 2026. https://r-libre.teluq.ca/4086/

[5] N. Dmitriev and E. Dynkin, *On the characteristic numbers of a stochastic matrix*, C. R. (Doklady) Acad. Sci. URSS (N.S.) **49** (1945), 159–162.

[6] N. Dmitriev and E. Dynkin, *On characteristic roots of stochastic matrices*, Izv. Akad. Nauk SSSR Ser. Mat. **10** (1946), 167–184.

[7] A. Harlev, C. R. Johnson and D. Lim, *The doubly stochastic single eigenvalue problem: a computational approach*, Exp. Math., doi:10.1080/10586458.2020.1727799; preprint arXiv:1908.03647.

[8] H. Ito, *A new statement about the theorem determining the region of eigenvalues of stochastic matrices*, Linear Algebra Appl. **267** (1997), 241–246.

[9] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292.

[10] F. I. Karpelevič, *On the characteristic roots of matrices with nonnegative elements*, Izv. Akad. Nauk SSSR Ser. Mat. **15** (1951), 361–383 (Russian); English translation in Amer. Math. Soc. Transl. (2) **140** (1988).

[11] B. Kim and J. Kim, *Conjectures about determining the regions of eigenvalues of stochastic and doubly stochastic matrices*, Linear Algebra Appl. **637** (2022), 157–174.

[12] J. Levick, R. Pereira and D. W. Kribs, *The four-dimensional Perfect–Mirsky conjecture*, Proc. Amer. Math. Soc. **143** (2015), no. 5, 1951–1956.

[13] J. Mashreghi and R. Rivard, *On a conjecture about the eigenvalues of doubly stochastic matrices*, Linear Multilinear Algebra **55** (2007), no. 5, 491–498.

[14] D. N. Munger, A. L. Nickerson and P. Paparella, *Demystifying the Karpelevič theorem*, Linear Algebra Appl. **702** (2024), 46–62.

[15] H. Perfect and L. Mirsky, *Spectral properties of doubly-stochastic matrices*, Monatsh. Math. **69** (1965), 35–57.

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